

UNIT 01

INTRODUCTION TO DATA SCIENCE AND MACHINE LEARNING

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Syllabus coverage:

Fundamentals of Artificial Intelligence • Need and Applications of Data Science • Data Mining • Data Preparation • Machine Learning • Types and Applications of Machine Learning

Learn the concept → Understand the example → Revise the key point

A. How to Use These Notes

- **Before class:** read the headings and study the diagrams.
- **During class:** add the teacher's examples in the blank space beside each topic.
- **After class:** use the “Key Point”, “Memory Aid” and comparison tables for revision.
- **Before examination:** attempt the MCQs, short-answer and long-answer questions without looking at the notes.

STUDY RULE

Do not only memorize definitions. Always connect each concept with one real-life example and one suitable application.

B. Learning Outcomes

- Define Artificial Intelligence, Data Science, Data Mining and Machine Learning.
- Explain why organizations need Data Science and where it is applied.
- Describe the stages of data preparation and the Data Science life cycle.
- Differentiate supervised, unsupervised, semi-supervised and reinforcement learning.
- Select a suitable learning type for a simple real-world problem.
- Identify basic ethical issues such as privacy, bias and transparency.

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1. Data, Information and Intelligence

1.1 What is Data?

Data consists of raw facts, observations or measurements collected about people, objects, events or processes. Data may be a number, word, image, sound, video, location, sensor reading or transaction record.

EXAMPLE

“78” is data. When we know that 78 is a student’s examination mark, it becomes meaningful information.

Term	Meaning	Example
Data	Raw fact without complete context	78, 92%, “CSE”, 30 July
Information	Processed or organized data with meaning	A student scored 78 marks
Knowledge	Understanding gained by interpreting information	The student performs well but needs improvement in one topic
Decision	An action based on knowledge	Provide targeted practice and feedback

1.2 Forms of Data

Form	Description	Examples
Structured	Organized in rows and columns with a fixed schema	Database table, spreadsheet, marksheet
Semi-structured	Has tags or keys but not a strict table format	JSON, XML, email headers
Unstructured	Has no predefined tabular structure	Images, audio, video, documents, social posts

MEMORY AID

Structured = table; Semi-structured = tagged; Unstructured = free-form.

1.3 What is Intelligence?

Intelligence is the ability to perceive, learn, reason, solve problems, communicate and take suitable action. In computing, these abilities are represented using rules, search, optimization, statistical models and learning algorithms.

- **Perception:** understanding images, sound, text or sensor input.
- **Reasoning:** drawing conclusions from facts and rules.
- **Learning:** improving performance from experience or data.
- **Planning:** choosing steps to achieve a goal.
- **Communication:** understanding and generating human language.

2. Fundamentals of Artificial Intelligence

2.1 Definition

Artificial Intelligence (AI) is a branch of computer science concerned with creating systems that can perform tasks normally associated with human intelligence, such as recognizing patterns, understanding language, solving problems and making decisions.

KEY POINT

AI is the broad field. Machine Learning is one important approach used to create AI systems.

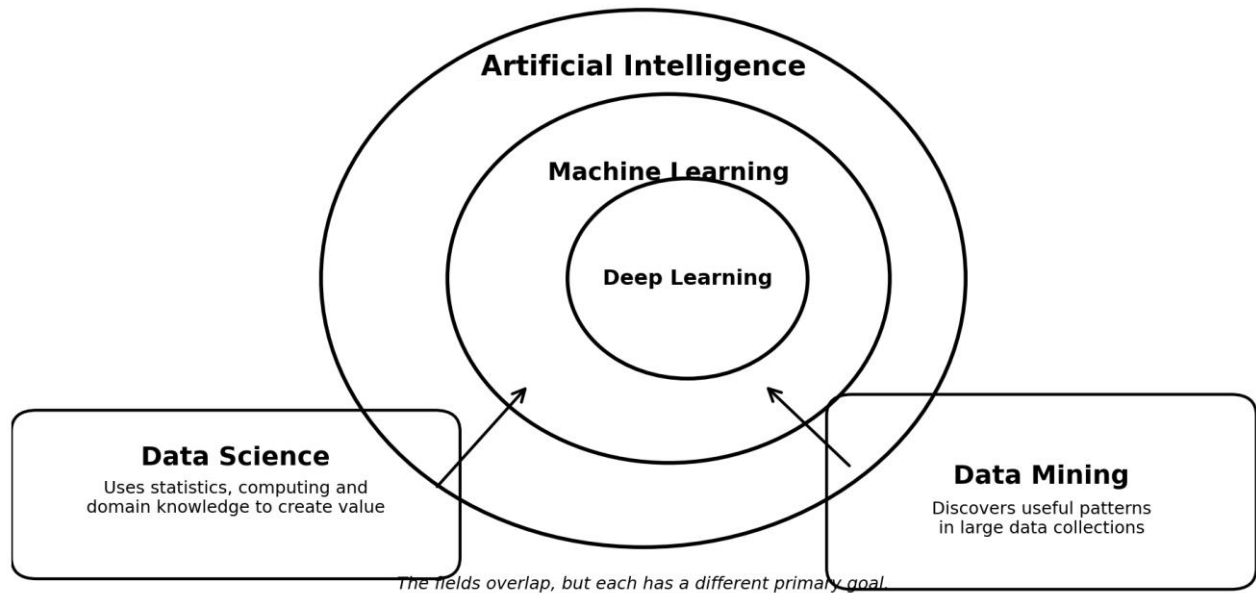


Figure 1: Relationship among AI, Machine Learning, Deep Learning, Data Science and Data Mining.

2.2 Main Capabilities of AI

Capability	What the system does	Example
Perception	Interprets visual, audio or sensor input	Face recognition, speech recognition
Knowledge representation	Stores facts and relationships in usable form	Knowledge graph, expert system
Reasoning	Uses rules or evidence to reach a conclusion	Medical decision support
Learning	Improves using data or feedback	Spam filter
Planning and action	Chooses a sequence of actions	Robot navigation
Natural language processing	Understands or produces language	Chatbot, translation

2.3 Narrow AI and General AI

Type	Meaning	Current status / example
Narrow AI	Designed for one limited task or domain	Common today: recommendation, face recognition, navigation
General AI	Hypothetical system with broad human-like ability across many tasks	Not currently available as a demonstrated general-purpose human-level system

EXAM TIP

Most systems used in daily life are Narrow AI. A system can be excellent at one task but unable to perform unrelated tasks.

2.4 Major Approaches in AI

- **Rule-based systems:** use IF–THEN rules prepared by experts.
- **Search and optimization:** explore possible solutions and select a good one.
- **Machine Learning:** learn patterns from data.
- **Deep Learning:** use multi-layer neural networks, especially for images, text and speech.
- **Hybrid systems:** combine rules, learning and human knowledge.

2.5 AI Problem-Solving Cycle

1. Receive input or observe the environment.
2. Represent the problem in a usable form.
3. Apply rules, search, optimization or a learned model.
4. Select a decision, output or action.
5. Use feedback to improve future performance.

2.6 Applications and Limitations

Area	Application
Education	Adaptive learning, automated feedback, student-support analytics
Healthcare	Image analysis, risk estimation, hospital resource planning
Banking	Fraud detection, credit-risk support, customer service
Transport	Route planning, traffic prediction, driver assistance
Agriculture	Crop monitoring, disease detection, yield estimation
Industry	Predictive maintenance, quality inspection, robotics

CAUTION

AI can make confident mistakes. Its output depends on data quality, design, objective and operating conditions. Human oversight remains important.

3. Need and Applications of Data Science

3.1 Definition

Data Science is an interdisciplinary field that uses data, statistics, computing, visualization and domain knowledge to understand problems, discover patterns, make predictions and support decisions.

Foundation	Role in Data Science
Statistics and Mathematics	Summarize data, estimate uncertainty and test relationships
Programming and Computing	Collect, clean, process and model data efficiently
Domain Knowledge	Interpret results correctly in the real application area
Communication and Visualization	Explain findings through charts, reports and dashboards

FORMULA

Data Science = Data + Methods + Computing + Domain Knowledge + Communication

3.2 Why is Data Science Needed?

- Very large amounts of digital data are generated by websites, mobile apps, sensors, machines and transactions.
- Manual examination of all records is slow and may miss important patterns.
- Organizations need evidence-based decisions rather than only intuition.
- Prediction helps plan demand, resources, risk and maintenance.
- Personalization improves services for different users.
- Monitoring helps detect unusual events, fraud and changing trends.

Typical Data Science Life Cycle

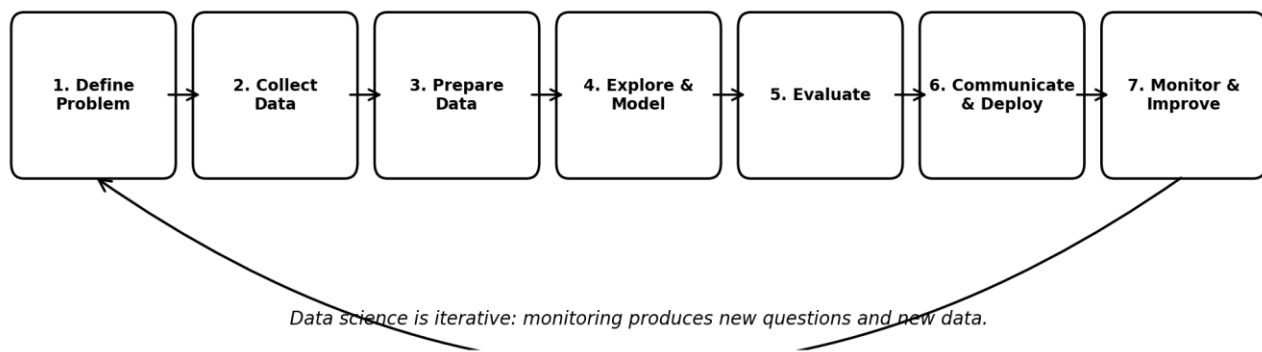


Figure 2: Typical Data Science life cycle. The process is iterative, not strictly one-way.

3.3 Data Science Life Cycle

Stage	Main question	Typical work
1. Define problem	What decision or question matters?	Convert a broad issue into a measurable objective
2. Collect data	What evidence is available?	Databases, files, APIs, surveys, sensors
3. Prepare data	Is the data reliable and usable?	Clean, integrate, transform and engineer features
4. Explore and model	What patterns or predictions are possible?	Statistics, visualization and ML algorithms
5. Evaluate	Is the result accurate, useful and fair?	Metrics, validation, error analysis
6. Communicate / deploy	How will people use the result?	Report, dashboard, application or workflow
7. Monitor	Does performance remain acceptable?	Track drift, errors and impact

3.4 Four Types of Analytical Questions

Type	Question	Example
Descriptive	What happened?	How many students passed?
Diagnostic	Why did it happen?	Why did pass percentage fall?
Predictive	What is likely to happen?	Which students may need support?
Prescriptive	What action should be taken?	Which intervention may improve results?

3.5 Applications of Data Science

Domain	Examples
Business and Retail	Customer segmentation, demand forecasting, recommendation
Education	Result analysis, attendance patterns, personalized learning
Healthcare	Disease-risk analysis, patient-flow planning, medical research
Government	Public-service planning, transport analysis, resource allocation
Science and Environment	Climate analysis, biodiversity monitoring, experiment data
Cybersecurity	Anomaly detection, suspicious-login analysis, threat monitoring

4. Data Mining

4.1 Meaning

Data Mining is the process of discovering useful patterns, relationships, trends and unusual cases in large data collections. It is an important stage within the broader Data Science process.

KEY POINT

Data Science covers the complete problem-solving life cycle. Data Mining mainly focuses on discovering patterns from prepared data.

4.2 Knowledge Discovery in Databases (KDD)

1. Selection: choose relevant data.
2. Cleaning: correct errors and inconsistencies.
3. Transformation: convert data into a suitable form.
4. Data mining: apply methods to discover patterns.
5. Evaluation: check whether patterns are valid and useful.
6. Presentation: communicate the discovered knowledge.

4.3 Common Data Mining Tasks

Task	Purpose	Example
Classification	Assign records to known categories	Spam / not spam
Regression	Predict a numeric value	Future sales
Clustering	Group similar records without labels	Customer groups
Association	Find items or events that occur together	Products bought together
Anomaly detection	Find rare or unusual cases	Suspicious transaction
Summarization	Create a compact description of data	Monthly performance profile

4.4 Association Rule Example

Suppose a shop observes that customers who buy bread and butter also often buy milk. A rule may be written as: {Bread, Butter} → {Milk}. The rule shows co-occurrence; it does not prove that bread causes milk purchase.

- **Support:** how frequently the item combination appears in all transactions.
- **Confidence:** how often the right-hand item appears when the left-hand items appear.

Field	Data Science	Data Mining	Machine Learning
Scope	Complete data-to-decision process	Pattern discovery from data	Algorithms that learn from data or feedback
Main output	Insight, report, model or deployed solution	Rules, groups, trends or anomalies	Predictive or decision-making model
Includes preparation?	Yes, as a major stage	Usually uses selected and prepared data	Required before training but not the complete field

5. Data Preparation

5.1 Meaning and Importance

Data preparation is the process of converting raw data into a clean, consistent and analysis-ready form. In real projects, this stage often requires more effort than model training because poor data produces unreliable results.

MEMORY AID Garbage In, Garbage Out (GIGO): incorrect or biased input data leads to incorrect or biased output.

Data Preparation Pipeline



Goal: improve data quality without leaking information from the future or test set.

Figure 3: Main stages in a data preparation pipeline.

5.2 Data Quality Dimensions

Dimension	Question
Accuracy	Does the value correctly represent reality?
Completeness	Are important values present?
Consistency	Do formats, labels and units agree across sources?
Validity	Does the value follow allowed rules or ranges?

Dimension	Question
Uniqueness	Are duplicate records avoided?
Timeliness	Is the data recent enough for the purpose?

5.3 Main Data Preparation Operations

Operation	What it means	Examples
Cleaning	Correct or handle data-quality problems	Missing values, duplicates, spelling errors, invalid ranges
Integration	Combine data from multiple sources	Join attendance and marks using student_id
Transformation	Change representation or scale	Date conversion, normalization, category encoding
Feature engineering	Create useful input variables	Attendance percentage, average test score
Feature selection	Keep relevant variables	Remove ID columns or redundant features
Reduction	Reduce size or complexity	Sampling, aggregation, dimensionality reduction

5.4 Handling Common Problems

- **Missing values:** remove the record only when justified, fill using a suitable value, or keep an explicit “unknown” category.
- **Duplicates:** identify repeated records and retain the authoritative record.
- **Inconsistent labels:** standardize entries such as “CSE”, “Computer Science” and “computer science”.
- **Outliers:** investigate extreme values before removing them; an outlier may be an error or a genuine rare event.
- **Invalid values:** apply domain rules, such as marks between 0 and 100.
- **Incorrect data types:** convert text dates, numeric strings and Boolean values into suitable types.

CAUTION

Never remove an outlier only because it is unusual. First determine whether it is an error, a valid rare case or an important event.

5.5 Feature, Label and Dataset Split

Term	Meaning	Student example
Feature (input)	Variable used to make a prediction	Attendance, internal marks, assignment score
Label / target (output)	Value the model should predict	Pass / fail
Training set	Data used to learn the model	Earlier student records used for learning
Validation set	Data used to tune choices	Used to compare settings or models
Test set	Unseen data used for final evaluation	Held-out records not used in training

EXAM TIP

Data leakage occurs when information unavailable at prediction time enters the training process. It gives unrealistically high performance.

5.6 Worked Cleaning Example

Raw record	Problem	Possible action
Course = “CSE”, “Computer Science”, “cse”	Inconsistent category labels	Map all to one standard label
Attendance = 112%	Invalid range	Verify source and correct; do not silently accept
Marks = “Absent” in numeric column	Mixed data type	Represent absence according to institutional policy
Same student appears twice	Duplicate	Use student_id and verify the authoritative record
One score is blank	Missing value	Investigate reason; choose a justified treatment

6. Machine Learning Fundamentals

6.1 Definition

Machine Learning (ML) is a method of building systems that learn patterns from data or feedback and improve their performance on a task without requiring every decision rule to be written manually.

Traditional Programming	Machine Learning
Rules + Data → Output	Data + Correct Outputs → Learned Model
Developer explicitly writes decision logic	Algorithm estimates decision logic from examples
Suitable when rules are clear and stable	Useful when patterns are complex or too numerous to write manually

EXAMPLE

Instead of writing thousands of rules for spam email, an ML model learns from examples labeled “spam” and “not spam”.

6.2 Essential Terms

Term	Meaning
Dataset	A collection of records used for analysis or learning
Sample / instance	One record or observation
Feature	An input variable describing a sample
Label / target	The correct output to be predicted
Algorithm	A method used to learn from data
Model	The learned pattern or function produced by training
Training	Process of learning model parameters from data
Prediction / inference	Using the trained model on new input
Metric	A numerical measure of performance

6.3 General ML Workflow

1. Define the task and success measure.
2. Collect relevant and ethically usable data.
3. Prepare the data and create features.
4. Split data into training, validation and test sets.
5. Choose a simple baseline and candidate algorithms.
6. Train the model.
7. Evaluate using suitable metrics and error analysis.
8. Check fairness, privacy, robustness and practical usefulness.
9. Deploy with appropriate human oversight.
10. Monitor performance and update when conditions change.

6.4 Generalization, Underfitting and Overfitting

Concept	Meaning	Result
Generalization	Performs well on new, unseen data	Desired behavior
Underfitting	Model is too simple or insufficiently trained	Poor on training and test data
Overfitting	Model memorizes training details and noise	Good training result but poor test result

MEMORY AID

Underfitting = model learns too little. Overfitting = model memorizes too much. Generalization = model learns the useful pattern.

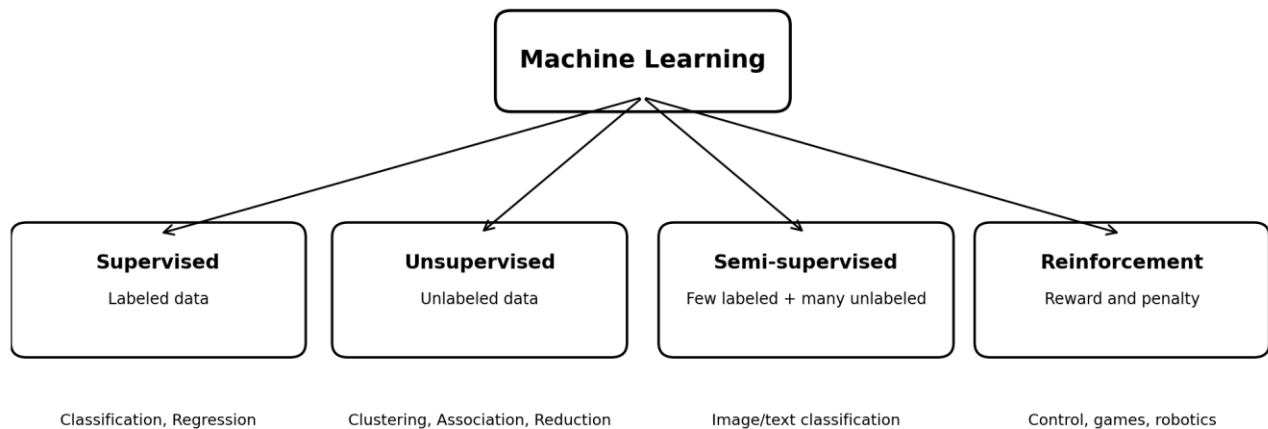
6.5 Basic Evaluation Measures

Task	Common measures	Simple meaning
Classification	Accuracy, precision, recall, F1-score	How correctly categories are predicted
Regression	MAE, MSE, RMSE, R^2	How close predicted numbers are to actual values
Clustering	Silhouette score and practical interpretation	How meaningful and separated the groups are

EXAM TIP

Accuracy alone may be misleading when one class is rare. Example: fraud, disease or at-risk students.

7. Types of Machine Learning



Choose the learning type by asking: What feedback or labels are available?

Figure 4: Major types of Machine Learning based on the feedback available to the learner.

7.1 Supervised Learning

Supervised learning uses labeled examples. Each training sample contains input features and the correct output. The model learns a mapping from inputs to outputs.

Task	Output	Examples
Classification	Category or class	Spam/not spam, pass/fail, disease present/absent
Regression	Continuous numeric value	House price, temperature, sales, delivery time

MEMORY AID Classification = class/category. Regression = real numeric value.

7.2 Unsupervised Learning

Unsupervised learning works with data that has no known target label. The system discovers hidden structure, groups, associations or compact representations.

Task	Purpose	Example
Clustering	Group similar samples	Student learning-profile groups
Association	Find events or items that occur together	Market basket analysis
Dimensionality reduction	Represent many variables using fewer dimensions	Visualization and compression
Anomaly detection	Identify unusual patterns	Network intrusion or fraud review

7.3 Semi-supervised Learning

Semi-supervised learning combines a small amount of labeled data with a larger amount of unlabeled data. It is useful when raw data is easy to collect but expert labeling is costly.

- Medical images labeled by specialists.
- Large image collections with labels for only a small subset.
- Speech or text data where manual annotation takes time.

7.4 Reinforcement Learning

In reinforcement learning, an agent interacts with an environment, takes actions and receives rewards or penalties. It learns a policy that aims to maximize long-term reward.

Element	Meaning	Example in a game
Agent	Learner or decision maker	Game-playing program
Environment	World in which the agent acts	Game board
State	Current situation	Current board position
Action	Choice made by the agent	Move a piece
Reward	Feedback after an action	Points, win or loss
Policy	Strategy for choosing actions	Rule learned for selecting moves

Learning type	Training information	Main goal	Example
Supervised	Labeled data	Predict known output	Email classification
Unsupervised	Unlabeled data	Discover structure	Customer clustering
Semi-supervised	Few labels + many unlabeled samples	Improve learning with limited labels	Medical image classification
Reinforcement	Rewards from interaction	Learn a sequence of actions	Robot navigation

8. Applications and Problem Selection

8.1 Match the Problem with the Learning Type

Problem statement	Expected output	Suitable type
Predict whether an email is spam	Spam / not spam	Supervised classification
Estimate next month's electricity demand	Numeric demand value	Supervised regression
Group customers by buying behavior	Natural customer groups	Unsupervised clustering
Find products commonly bought together	Association rules	Unsupervised association mining
Use a few labeled X-rays and many unlabeled X-rays	Disease category	Semi-supervised learning
Teach a robot to reach a destination through trial and feedback	Sequence of actions	Reinforcement learning

8.2 Common Applications

Application	Possible data	Possible ML task
Recommendation system	Clicks, ratings, purchases	Ranking / prediction / clustering
Fraud detection	Transaction time, amount, location	Classification or anomaly detection
Predictive maintenance	Sensor readings and failure history	Classification or regression
Image recognition	Labeled images	Classification
Demand forecasting	Historical sales, season, price	Regression / time-series prediction
Language translation	Source and target sentences	Sequence learning

SELECTION RULE

First identify the required output: category, number, group, association or sequence of actions. Then choose the learning type.

8.3 AI vs ML vs Data Science: One Example

Consider an online shopping recommendation feature:

- **Data Science:** defines the business question, prepares customer data, evaluates impact and communicates results.
- **Data Mining:** discovers products frequently purchased together or customer segments.
- **Machine Learning:** trains a model to predict which products a user may prefer.
- **Artificial Intelligence:** is the broader goal of providing intelligent, adaptive recommendations.

9. Integrated Case Study: Student Support Analytics

9.1 Problem Statement

A college wants to identify students who may need academic support before the final examination. The purpose is to provide help early, not to punish or permanently label students.

9.2 Possible Data

- Attendance records
- Internal examination marks
- Assignment completion
- Practical performance
- Previous semester results
- Support sessions already provided

9.3 Data Preparation

1. Join different tables using student_id.
2. Remove confirmed duplicate records.
3. Check attendance and marks for impossible values.
4. Standardize course and semester labels.
5. Create features such as attendance percentage and average internal score.
6. Exclude irrelevant or unfair variables.
7. Split data so that future information does not leak into training.

9.4 Possible ML Formulations

Question	Formulation
Will the student require support?	Binary classification
What final score is expected?	Regression
What learning profiles exist?	Clustering
Which attendance and assessment patterns occur together?	Association analysis

9.5 Evaluation and Responsible Use

- Check how many genuinely at-risk students are identified (recall).
- Review false alarms to avoid unnecessary anxiety or stigma.
- Allow teachers to verify records and override incorrect output.
- Protect student data and limit access.
- Evaluate whether the support intervention improves outcomes.

KEY POINT

A prediction should support a human decision; it should not automatically determine a student's future.

10. Responsible Data Science and AI

10.1 Main Concerns

Concern	Meaning	Good practice
Privacy	Personal data may be exposed or misused	Collect minimum necessary data and control access
Bias and fairness	Some groups may receive systematically worse results	Check data representation and compare errors across groups
Transparency	Users may not understand how a result was produced	Document data, methods, limitations and intended use
Security	Data or models may be attacked or stolen	Use authentication, encryption and monitoring
Accountability	Responsibility may become unclear	Assign human owners and review processes
Reliability	Performance may fall when conditions change	Monitor errors and data drift

10.2 Responsible Development Checklist

- Is the problem worth solving with data or AI?
- Do we have permission and a valid purpose for using the data?
- Is the data accurate, representative and sufficiently recent?
- Which people may be helped or harmed by errors?
- Can a human review or correct the output?
- How will performance, fairness and security be monitored?

CAUTION

A highly accurate model may still be unsuitable if it violates privacy, creates unfair treatment or cannot be used safely.

11. Quick Revision and Glossary

11.1 One-Page Concept Summary

Concept	One-line revision
Artificial Intelligence	Broad field of creating systems that perform intelligent tasks.
Data Science	Complete process of using data to understand problems and support decisions.
Data Mining	Discovery of useful patterns, trends and anomalies in data.
Data Preparation	Cleaning, integrating and transforming raw data into usable form.
Machine Learning	Algorithms that learn patterns from data or feedback.
Supervised Learning	Learns from labeled input-output examples.
Unsupervised Learning	Discovers structure in unlabeled data.
Semi-supervised Learning	Uses few labeled and many unlabeled examples.
Reinforcement Learning	Learns actions using rewards and penalties.
Classification	Predicts a category.
Regression	Predicts a numeric value.
Clustering	Groups similar samples without known labels.

11.2 Important Differences

Pair	Difference
AI vs ML	AI is the broad goal/field; ML is a data-driven approach within AI.
Data Science vs Data Mining	Data Science is the full life cycle; Data Mining focuses on pattern discovery.
Classification vs Regression	Classification predicts categories; regression predicts continuous numbers.
Supervised vs Unsupervised	Supervised uses labels; unsupervised has no known target labels.
Training vs Testing	Training teaches the model; testing measures performance on unseen data.
Feature vs Label	Feature is an input; label is the target output.

11.3 Glossary

Term	Meaning
Algorithm	A step-by-step computational method.
Anomaly	An unusual record or pattern that differs strongly from normal data.
Bias	A systematic tendency that may produce unfair or inaccurate results.
Data leakage	Use of information during training that would not be available at prediction time.
Deep Learning	ML based on multi-layer neural networks.
Feature engineering	Creating useful input variables from existing data.

Term	Meaning
Inference	Using a trained model to make a prediction.
Model	A learned mathematical representation used for prediction or decision-making.
Outlier	An observation far from most other observations.
Overfitting	Learning training details so closely that new-data performance becomes poor.

12. Self-Assessment

12.1 Multiple-Choice Questions

- Which term refers to raw facts and observations?**
a) Model b) Data c) Knowledge d) Decision
- Most AI systems used today are:**
a) General AI b) Narrow AI c) Human intelligence d) None
- Which field covers the complete data-to-decision process?**
a) Data Science b) Data entry c) Networking d) Operating system
- Finding products that occur together is mainly:**
a) Regression b) Association mining c) Compilation d) Encryption
- Which is a data-quality dimension?**
a) Accuracy b) Animation c) Inheritance d) Recursion
- Predicting pass/fail is:**
a) Regression b) Classification c) Clustering d) Compression
- Predicting house price is:**
a) Regression b) Classification c) Association d) Reinforcement
- Grouping customers without labels is:**
a) Supervised learning b) Clustering c) Regression d) Rule writing
- Which learning type uses rewards and penalties?**
a) Reinforcement b) Supervised c) Unsupervised d) Semi-structured
- A model that performs well on training data but poorly on new data is:**
a) Generalized b) Overfitted c) Encrypted d) Normalized
- Information unavailable at prediction time entering training is:**
a) Data leakage b) Sampling c) Visualization d) Clustering
- Which set is used for final evaluation on unseen data?**
a) Training set b) Test set c) Duplicate set d) Rule set

12.2 Short-Answer Questions

- Define Artificial Intelligence with one example.
- What is Data Science? Write any four foundations of Data Science.
- Why is data preparation important?
- Differentiate structured and unstructured data.
- Write the stages of the Data Science life cycle.
- What is Data Mining? List four common tasks.
- Differentiate classification and regression.

8. What is overfitting?
9. Explain semi-supervised learning with one use case.
10. List four responsible-AI concerns.

12.3 Long-Answer Questions

1. Explain the fundamentals, capabilities, applications and limitations of Artificial Intelligence.
2. Explain the need for Data Science and describe its life cycle with a suitable example.
3. Describe the complete data preparation process and common data-quality problems.
4. Explain supervised, unsupervised, semi-supervised and reinforcement learning with applications.
5. Develop a simple Data Science plan for identifying students who may require academic support.

12.4 Classroom / Homework Activities

Activity	Task
Data audit	Take a small marksheet and identify missing values, duplicates and invalid values.
Problem matching	Write five real-life problems and identify classification, regression, clustering or association.
Life-cycle map	Choose one college problem and describe all stages from problem definition to monitoring.
Ethics discussion	Explain one benefit and one possible harm of using AI in education.

12.5 MCQ Answer Key

1-b, 2-b, 3-a, 4-b, 5-a, 6-b, 7-a, 8-b, 9-a, 10-b, 11-a, 12-b

FINAL REVISION	For every topic, remember: definition → main features → example → application → limitation or caution.
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